

what is difference between ISO18000-6C and ISO18000-63 Protocol?

I'll compare them from the RFID reader/module point of view, because that is usually where the confusion happens: many vendors say "ISO18000-6C" while newer documents say "ISO18000-63."

Thought for 12s

Main difference: almost no practical difference.

ISO/IEC 18000-6C is the older name for the UHF RFID air-interface protocol commonly known as **EPC Class-1 Gen-2 / EPC Gen2**.

ISO/IEC 18000-63 is the newer ISO naming format for basically the same UHF RFID air-interface standard. RAIN RFID also describes RAIN systems as using **ISO/IEC 18000-63**, also known as **GS1 UHF Gen2**. The RAIN... +1

Simple comparison

Item	ISO18000-6C	ISO18000-63
Meaning	Old name	Newer ISO name
RFID type	UHF RFID	UHF RFID
Frequency	860–960 MHz range	860–960 MHz range
Common name	EPC Class-1 Gen-2 / EPC Gen2	EPC Gen2 / GS1 UHF Gen2
Reader/tag compatibility	Same family	Same family
Used for	Inventory, logistics, asset tracking, warehouse, access, etc.	Same
Practical difference for module selection	Usually none	Usually none

ISO’s own page says **ISO/IEC 18000-63** defines the air interface for RFID devices operating in the **860 MHz to 960 MHz** band for item management. ISO

Important point for your RS2000/R2000 UHF reader module

When a seller says:

Protocol: EPC C1G2 / ISO18000-6C / ISO18000-63

It usually means:

The module supports standard UHF RFID Gen2 tags.

So, for purchasing, you can treat them as compatible in most cases.

Best wording to ask supplier

You can ask like this:

Please confirm that the module supports EPC Class-1 Gen-2 / ISO/IEC 18000-63, formerly ISO/IEC 18000-6C, including inventory, read, write, lock, kill, select, and access password commands.

One small caution

The protocol name does **not** guarantee everything. You still need to confirm:

Need to confirm	Why
Frequency region	US, EU, China, etc. use different allowed bands
Output power	Affects reading distance
Command protocol	MCU talks to reader module using vendor UART/USB protocol, not raw ISO air protocol
Tag memory support	EPC, TID, User, Reserved memory
SDK / documentation	Needed for STM32 integration

In short: **ISO18000-6C** is the old/common name; **ISO18000-63** is the newer/current ISO name. For normal UHF RFID Gen2 reader/tag compatibility, they refer to the same protocol family.

generate the datasheet for each protocol

I'll make this as **practical engineering datasheets**, not the full official ISO text. The official standards are copyrighted, so I'll summarize the useful implementation-level information: scope, frequency, modulation, memory, commands, and what it means for an RFID reader module.

Thought for 6s

Below are **datasheet-style summaries** for both protocol names.

Important: **ISO18000-6C** and **ISO18000-63** are not really two different protocols. For UHF RFID, **ISO/IEC 18000-6C** is the older/common name, and **ISO/IEC 18000-63** is the newer ISO naming for **860–960 MHz Type C**, also known as **EPC Class-1 Gen-2 / GS1 UHF Gen2 / RAIN RFID**. ISO describes ISO/IEC 18000-63 as the air interface for RFID devices operating in the **860 MHz to 960 MHz** band for item management. ISO +1

Datasheet 1 — ISO/IEC 18000-6C Protocol

1. General Information

Item	Description
Protocol name	ISO/IEC 18000-6C
Common name	EPC Class-1 Generation-2, EPC C1G2, EPC Gen2
Newer ISO name	ISO/IEC 18000-63
RFID type	Passive UHF RFID
Frequency band	860 MHz to 960 MHz
Application	Item management, inventory, logistics, warehouse, asset tracking
Communication type	Reader-to-tag and tag-to-reader air interface

Item	Description
Tag power	Passive tag powered by reader RF field

2. System Components

Component	Description
Interrogator / Reader	Sends RF commands and powers passive tags
Tag / Transponder	Passive RFID tag containing EPC/TID/User memory
Antenna	Transmits and receives UHF RF signal
Host MCU / PC	Controls reader module through UART, USB, SPI, TCP, etc.

3. Frequency and Region

Region	Typical UHF RFID band
US / FCC	902–928 MHz
Europe / ETSI	865–868 MHz
China	Usually around 840–845 MHz and/or 920–925 MHz depending on regulation

The protocol defines the air interface, but the **legal frequency band and output power depend on the country.**

4. Tag Memory Banks

Memory Bank	Bank No.	Description
Reserved	00	Kill password and Access password
EPC	01	EPC code, PC bits, CRC
TID	10	Tag identifier / manufacturer information

Memory Bank	Bank No.	Description
User	11	Optional user data memory

5. Main Commands

Command	Purpose
Query	Start inventory round
QueryRep	Continue inventory round
QueryAdjust	Adjust inventory Q value
ACK	Acknowledge selected tag RN16
NAK	Force tag to return to arbitrate state
Req_RN	Request new handle from tag
Read	Read tag memory
Write	Write tag memory
Kill	Permanently disable tag
Lock	Lock or unlock memory/password access
Access	Move tag into secured state
Select	Select tag population before inventory
BlockWrite	Write multiple words, if supported
BlockErase	Erase multiple words, if supported

The GS1 Gen2 protocol specification lists commands such as **Req_RN**, **Read**, **Write**, **Lock**, **Kill**, **Access**, and **BlockWrite** as part of the Gen2 command set. GS1

6. Basic Inventory Flow



7. Practical Reader Module Meaning

For an RS2000/R2000 UHF RFID module, ISO18000-6C support usually means the module can:

Function	Supported meaning
Inventory tags	Detect multiple UHF tags
Read EPC	Read tag EPC memory
Read TID	Read unique tag/chip identifier
Read User memory	Read optional user memory
Write EPC/User	Write data to writable memory
Lock memory	Protect memory banks
Kill tag	Permanently disable tag, if password is set
Anti-collision	Read many tags in field

8. Important Note for STM32

Your STM32 usually does **not directly implement ISO18000-6C RF modulation.**

Usually:

STM32

↓ UART / USB / SPI

UHF RFID reader module

↓ RF air interface

ISO18000-6C / EPC Gen2 tag

So your STM32 code normally implements the **reader module serial command protocol**, not the raw ISO18000-6C air protocol.

Datasheet 2 — ISO/IEC 18000-63 Protocol

1. General Information

Item	Description
Protocol name	ISO/IEC 18000-63
Full meaning	Parameters for air interface communications at 860 MHz to 960 MHz Type C
Older/common name	ISO/IEC 18000-6C
Common industry name	EPC Class-1 Gen-2, GS1 UHF Gen2, RAIN RFID
RFID type	Passive UHF RFID
Frequency band	860 MHz to 960 MHz
Application	Item-level RFID, inventory, logistics, retail, warehouse, asset tracking
Compatibility goal	Interoperability between compliant readers and tags

RAIN RFID describes RAIN systems as using the **ISO/IEC 18000-63 air-interface protocol**, also known as **GS1 UHF Gen2**.

The RAIN Alliance

2. Relationship to ISO18000-6C

Item	ISO18000-6C	ISO18000-63
Naming	Older Part 6 Type C naming	Newer Part 63 naming

Item	ISO18000-6C	ISO18000-63
Frequency	860–960 MHz	860–960 MHz
Type	Type C	Type C
Common name	EPC Gen2	EPC Gen2 / GS1 UHF Gen2
Compatibility	Same protocol family	Same protocol family

So in most datasheets:

ISO18000-6C = ISO18000-63 = EPC C1G2 = UHF Gen2

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3. Air Interface Scope

ISO/IEC 18000-63 defines communication between:

Reader / Interrogator ↔ Passive UHF Tag

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It covers:

Area	Description
Physical layer	RF signaling between reader and tag
Tag inventory	Multi-tag anti-collision process
Tag access	Read, write, lock, kill operations
Memory access	EPC, TID, User, Reserved memory
Compatibility	Standard behavior for interoperable readers/tags

4. Memory Model

Memory	Description
Reserved memory	Stores Kill password and Access password
EPC memory	Stores EPC identifier and protocol control bits

Memory	Description
TID memory	Stores tag/chip identification data
User memory	Optional data storage area

5. Command Categories

Inventory Commands

Command	Description
Select	Filter/select target tag population
Query	Start inventory
QueryRep	Continue inventory slot
QueryAdjust	Modify Q parameter
ACK	Acknowledge tag response
NAK	End current tag interaction

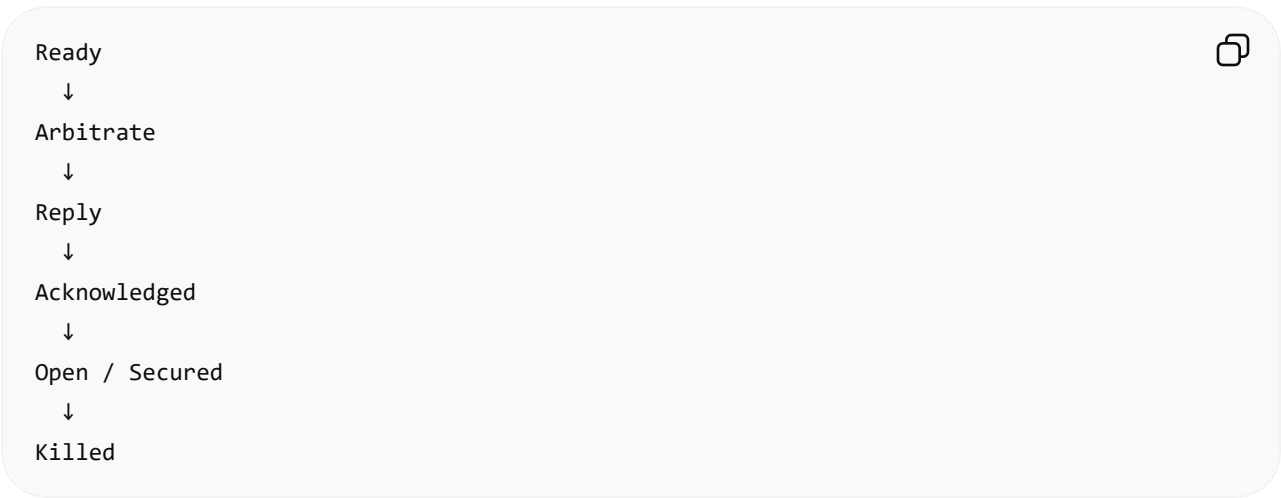
Access Commands

Command	Description
Req_RN	Request tag handle
Read	Read tag memory
Write	Write one word to tag memory
Lock	Lock/password-protect memory
Kill	Permanently disable tag
Access	Enter secured access state
BlockWrite	Write multiple words if supported

Command	Description
BlockErase	Erase multiple words if supported

6. Tag States

Typical tag state flow:

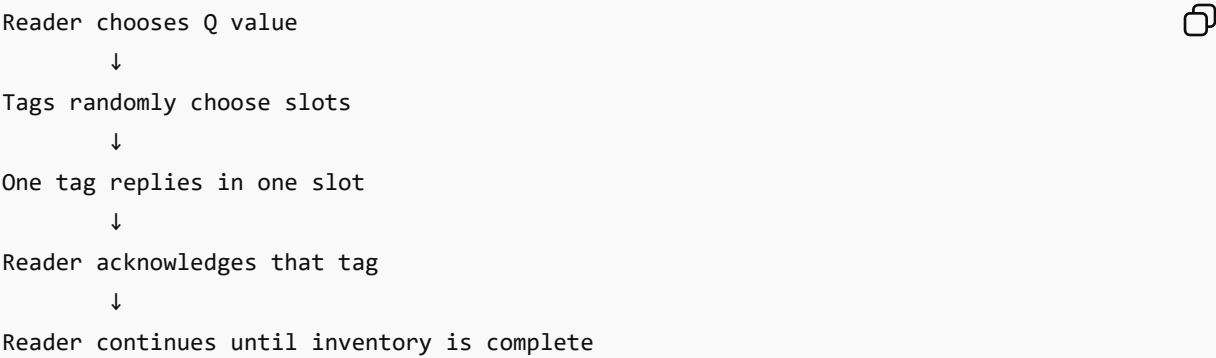


State	Meaning
Ready	Tag is powered and waiting
Arbitrate	Tag participates in anti-collision
Reply	Tag replies with RN16
Acknowledged	Tag has been selected by reader
Open	Tag can respond to some access commands
Secured	Tag has accepted access process/password
Killed	Tag is permanently disabled

7. Anti-Collision

ISO18000-63 / EPC Gen2 supports reading many passive tags by using a slotted anti-collision process.

Simplified:



8. Practical Reader Module Meaning

If a reader module says it supports ISO18000-63, it should normally support:

Feature	Meaning
Multi-tag inventory	Read many tags in antenna field
EPC read	Read normal EPC code
TID read	Read chip/manufacturer ID
User memory access	Read/write user memory if tag supports it
Password operation	Access/Kill password functions
Lock operation	Protect memory areas
Regional frequency setting	Configure FCC/ETSI/China/etc. band
Output power control	Set RF power level
Antenna control	Select antenna port if module has multiple ports

Combined Practical Datasheet for Supplier Request

You can send this to a supplier:

Required UHF RFID Protocol Support

The reader module must support:

1. Air Interface
 - EPC Class-1 Gen-2 / EPC C1G2
 - ISO/IEC 18000-6C
 - ISO/IEC 18000-63, 860 MHz to 960 MHz Type C
 - RAIN RFID / GS1 UHF Gen2 compatible
2. Tag Operations
 - Inventory / multi-tag anti-collision
 - Read EPC memory
 - Read TID memory
 - Read User memory
 - Write EPC memory
 - Write User memory
 - Lock memory
 - Kill tag
 - Access password support
 - Select/filter command support
3. Memory Banks
 - Reserved memory
 - EPC memory
 - TID memory
 - User memory
4. Region Support
 - FCC / US: 902-928 MHz
 - ETSI / EU: 865-868 MHz
 - China frequency option if available
 - Configurable frequency hopping table
5. Host Interface
 - UART or USB or SPI interface
 - Complete command protocol document
 - Frame format
 - Checksum rule
 - Inventory command frame
 - Read command frame
 - Write command frame
 - Lock command frame
 - Kill command frame
 - Response frame format
 - Error code list
6. Development Support
 - Datasheet
 - Pin definition
 - Electrical specification

- SDK or demo source code
- STM32 example if available

Very important for your STM32 project

For STM32F407, you should ask for **module command protocol**, not only ISO18000-6C / ISO18000-63.

Because ISO18000-63 is the **RF air protocol** between reader and tag.

But STM32 communicates like this:

STM32F407

↓ UART command frame

RS2000/R2000 reader module

↓ ISO18000-63 RF protocol

UHF RFID tag



So the document you really need from the supplier is usually called:

Document name	Needed?
Module datasheet	Yes
Pin definition	Yes
UART/USB command protocol	Very important
SDK manual	Very useful
EPC Gen2 / ISO18000-63 standard	Useful, but not enough for coding STM32
Demo source code	Very useful

Best sentence to supplier:

Please provide the RS2000/R2000 module UART command protocol document, including frame



